

Sorting Templates

All recursive sorting uses **half-open intervals** `[start, end)` throughout.
Inside merge and partition: `i` , `j` , `k` as scan/write pointers (consistent with binary search convention).

Quick Reference Table

#	Name	Description	Intuition	Time	Space	Key Implementation
912	Sort an Array	Sort an integer array. No built-in sort allowed.	Divide and conquer — either split-then-merge (merge sort) or partition-then-recurse (quick sort); both reach $O(n \log n)$.	$O(n \log n)$	$O(n)$ merge / $O(\log n)$ quick	Standard
215	Kth Largest Element in an Array	Return the k-th largest element without sorting the full array.	Quick select partitions around a pivot and only recurses into the side containing the target rank, so it averages $O(n)$ instead of fully sorting.	$O(n)$ avg	$O(\log n)$	Standard
973	K Closest Points to Origin	Return the k closest points to origin (0,0). Any order.	Same quick select, but partition on squared distance — once the first k slots are filled with the closest points, their internal order doesn't matter.	$O(n)$ avg	$O(\log n)$	Standard
148	Sort List	Sort a linked list in $O(n \log n)$ time, $O(\log n)$ space.	Merge sort fits linked lists — there's no random access for quick sort, but splitting at the middle and merging two sorted lists needs only pointer rewiring.	$O(n \log n)$	$O(\log n)$	Standard
179	Largest Number	Given a list of non-negative integers, arrange them to form the largest number.	Order two numbers by which concatenation is bigger (<code>b+a</code> vs <code>a+b</code>) — this pairwise rule sorts the whole list into the largest possible string.	$O(n \log n)$	$O(n)$	Standard
315	Count of Smaller Numbers After Self	For each element, count how many elements to its right are smaller.	During a merge, whenever a left-half element is placed, every right-half element already emitted is both smaller and to its right — so the merge counts the smaller-after-self pairs for free.	$O(n \log n)$	$O(n)$	Standard
56	Merge Intervals	Merge all overlapping intervals. Return array of non-overlapping intervals.	After sorting by start, overlaps can only be with the most recent interval, so one scan either extends its end or appends a new one.	$O(n \log n)$	$O(n)$	Standard
88	Merge Sorted Array	Given two sorted arrays <code>nums1</code> (with capacity for $m+n$) and <code>nums2</code> , merge <code>nums2</code> into <code>nums1</code> in-place.		$O(m+n)$	$O(1)$	Fill from the END backwards using three pointers <code>i</code> (end of <code>nums1</code> valid data), <code>j</code> (end of <code>nums2</code>), <code>k</code> (end of merged result). This avoids the need to shift elements.
327	Count of Range Sum	Given an integer array, count the number of range sums <code>sum(i, j)</code> that lie within <code>[lower, upper]</code> (inclusive).		$O(n \log n)$	$O(n)$	Build prefix sum array. Then use modified merge sort — during the merge phase, for each element in the left half, use two sliding pointers <code>j</code> and <code>k</code> into the right half to count valid prefix sums (those where <code>arr[k] - arr[i]</code> falls in range). This gives $O(n \log n)$ vs $O(n^2)$ brute force.
164	Maximum Gap	Given an unsorted array, return the maximum difference between successive elements in its sorted form. Must run in linear time.		$O(n)$	$O(n)$	bucket sort by pigeonhole principle. With n elements, the max gap is at least <code>ceil((max-min)/(n-1))</code> . Place elements into buckets of that size; the max gap spans between one bucket's max and the next non-empty bucket's min.

Template 1 — Merge Sort

```
// MENTAL MODEL: split in half until trivially sorted, then merge two sorted halves into one.
// WHEN: need stable sort, sort a linked list, or count cross-pairs (inversions) during the merge.
// Entry point
public void mergeSort(int[] nums) {
    mergeSort(nums, 0, nums.length, new int[nums.length]);
}

// [start, end) half-open
private void mergeSort(int[] nums, int start, int end, int[] temp) {
    if (end - start <= 1) return;          // base case: end - start <= 1 → 0 or 1 elements, already sorted
    int mid = start + (end - start) / 2;
    mergeSort(nums, start, mid, temp);    // sort [start, mid)
    mergeSort(nums, mid, end, temp);      // sort [mid, end)
    merge(nums, start, mid, end, temp);
}

// i scans left [start,mid), j scans right [mid,end), k writes to temp
private void merge(int[] nums, int start, int mid, int end, int[] temp) {
    int i = start, j = mid, k = start;
    while (i < mid || j < end) {
        if (j >= end || (i < mid && nums[i] <= nums[j])) {
            temp[k++] = nums[i++];
        } else {
            temp[k++] = nums[j++];
        }
    }
    for (i = start; i < end; i++) nums[i] = temp[i];
}
```

Template 2 — Quick Sort (3-way Dutch Flag Partition)

```
// MENTAL MODEL: pick a pivot, partition into <pivot | ==pivot | >pivot, recurse on the two ends only.
// WHEN: in-place sort with O(log n) stack; the ==pivot middle is already in final position.
// Entry point – no extra array needed (in-place)
public void quickSort(int[] nums) {
    quickSort(nums, 0, nums.length);
}

// [start, end) half-open
private void quickSort(int[] nums, int start, int end) {
    if (end - start <= 1) return;
    int pivot = nums[start + (end - start) / 2];
    int i = start, k = start, j = end - 1;
    // Invariant: [start,i) < pivot | [i,k) == pivot | [k,j) unknown | (j,end) > pivot
    while (k <= j) {
        if (nums[k] < pivot) {
            swap(nums, k++, i++);
        } else if (nums[k] == pivot) {
            k++;
        } else {
            swap(nums, k, j--);
        }
    }
    // After: [start,i) < pivot, [i,k) == pivot, [k,end) > pivot
    quickSort(nums, start, i); // recurse on < region
    quickSort(nums, k, end);   // recurse on > region
}

private void swap(int[] nums, int i, int j) {
    int t = nums[i]; nums[i] = nums[j]; nums[j] = t;
}
```

Template 3 — Quick Select (partial sort — k-th element)

Extends quick sort: after partition, only recurse into the side that contains `target` index.

```
// MENTAL MODEL: same partition as quick sort, but recurse into only the one side holding target → O(n) avg.
// WHEN: "k-th largest/smallest", "k closest/most frequent" – you need a position, not a full sort.
// Returns value at 0-indexed position target after sorting
private int quickSelect(int[] nums, int start, int end, int target) {
    int pivot = nums[start + (end - start) / 2];
    int i = start, k = start, j = end - 1;
    while (k <= j) {
        if (nums[k] < pivot) {
            swap(nums, k++, i++);
        } else if (nums[k] == pivot) {
            k++;
        } else {
            swap(nums, k, j--);
        }
    }
    // [start,i) < pivot, [i,k) == pivot, [k,end) > pivot
    if (target < i) {
        return quickSelect(nums, start, i, target);
    } else if (target < k) {
        return pivot; // target lands in == region
    } else {
        return quickSelect(nums, k, end, target);
    }
}
```

Template 4 — Counting Sort

$O(n + \text{range})$. Use when values are bounded integers.

```
// MENTAL MODEL: tally how many times each value occurs, then emit values in order — no comparisons.
// WHEN: integers in a small bounded range; beats  $O(n \log n)$  when  $\text{range} = O(n)$ .
public void countingSort(int[] arr) {
    if (arr.length == 0) return;
    int min = Arrays.stream(arr).min().getAsInt();
    int max = Arrays.stream(arr).max().getAsInt();
    int[] buckets = new int[max - min + 1];
    for (int x : arr) buckets[x - min]++;
    int i = 0;
    for (int v = 0; v < buckets.length; v++)
        while (buckets[v]-- > 0) arr[i++] = v + min;
}
```

#912 Sort an Array

Description: Sort an integer array. No built-in sort allowed.

Intuition: Divide and conquer — either split-then-merge (merge sort) or partition-then-recurse (quick sort); both reach $O(n \log n)$.

```
// Merge Sort version
class Solution {
    public int[] sortArray(int[] nums) {
        mergeSort(nums, 0, nums.length, new int[nums.length]);
        return nums;
    }
    private void mergeSort(int[] nums, int start, int end, int[] temp) {
        if (end - start <= 1) return;
        int mid = start + (end - start) / 2;
        mergeSort(nums, start, mid, temp);
        mergeSort(nums, mid, end, temp);
        merge(nums, start, mid, end, temp);
    }
    private void merge(int[] nums, int start, int mid, int end, int[] temp) {
        int i = start, j = mid, k = start;
        while (i < mid || j < end) {
            if (j >= end || (i < mid && nums[i] <= nums[j])) {
                temp[k++] = nums[i++];
            } else {
                temp[k++] = nums[j++];
            }
        }
        for (i = start; i < end; i++) nums[i] = temp[i];
    }
}

// Quick Sort version (in-place, O(log n) stack)
class Solution {
    public int[] sortArray(int[] nums) {
        quickSort(nums, 0, nums.length);
        return nums;
    }
    private void quickSort(int[] nums, int start, int end) {
        if (end - start <= 1) return;
        int pivot = nums[start + (end - start) / 2];
        int i = start, k = start, j = end - 1;
        while (k <= j) {
            if (nums[k] < pivot) {
                swap(nums, k++, i++);
            } else if (nums[k] == pivot) {
                k++;
            } else {
                swap(nums, k, j--);
            }
        }
        quickSort(nums, start, i);
        quickSort(nums, k, end);
    }
    private void swap(int[] nums, int i, int j) {
        int t = nums[i]; nums[i] = nums[j]; nums[j] = t;
    }
}
```

Time $O(n \log n)$ | **Space** $O(n)$ merge / $O(\log n)$ quick

#215 Kth Largest Element in an Array

Description: Return the k-th largest element without sorting the full array.

Intuition: Quick select partitions around a pivot and only recurses into the side containing the target rank, so it averages $O(n)$ instead of fully sorting.

Key: k-th largest = $(n - k)$ -th smallest (0-indexed). Quick select, recurse one side only.

```

class Solution {
    public int findKthLargest(int[] nums, int k) {
        return quickSelect(nums, 0, nums.length, nums.length - k);
    }
    private int quickSelect(int[] nums, int start, int end, int target) {
        int pivot = nums[start + (end - start) / 2];
        int i = start, k = start, j = end - 1;
        while (k <= j) {
            if (nums[k] < pivot) {
                swap(nums, k++, i++);
            } else if (nums[k] == pivot) {
                k++;
            } else {
                swap(nums, k, j--);
            }
        }
        if (target < i) {
            return quickSelect(nums, start, i, target);
        } else if (target < k) {
            return pivot;
        } else {
            return quickSelect(nums, k, end, target);
        }
    }
    private void swap(int[] nums, int i, int j) {
        int t = nums[i]; nums[i] = nums[j]; nums[j] = t;
    }
}

```

Time $O(n)$ avg | **Space** $O(\log n)$

#973 K Closest Points to Origin

Description: Return the k closest points to origin (0,0). Any order.

Intuition: Same quick select, but partition on squared distance — once the first k slots are filled with the closest points, their internal order doesn't matter.

Key: same quick select template; comparator = squared distance (avoid sqrt). After select, first k entries in `[0, k)` are the answer.

```

class Solution {
    public int[][] kClosest(int[][] pts, int k) {
        quickSelect(pts, 0, pts.length, k);
        return Arrays.copyOfRange(pts, 0, k);
    }
    private void quickSelect(int[][] pts, int start, int end, int k) {
        if (end - start <= k) return;
        long pivot = dist(pts[start + (end - start) / 2]);
        int i = start, current = start, j = end - 1;
        while (current <= j) {
            long d = dist(pts[current]);
            if (d < pivot) {
                swap(pts, current++, i++);
            } else if (d == pivot) {
                current++;
            } else {
                swap(pts, current, j--);
            }
        }
        // [start,i) < pivot, [i,current) == pivot, [current,end) > pivot
        int less = i - start, equal = current - i;
        if (k <= less) {
            quickSelect(pts, start, i, k);
        } else if (k <= less + equal) {
            return; // enough elements in [start,current)
        } else {
            quickSelect(pts, current, end, k - less - equal);
        }
    }
    private long dist(int[] p) { return (long) p[0]*p[0] + (long) p[1]*p[1]; }
    private void swap(int[][] pts, int i, int j) {
        int[] t = pts[i]; pts[i] = pts[j]; pts[j] = t;
    }
}

```

Time $O(n)$ avg | **Space** $O(\log n)$

#148 Sort List

Description: Sort a linked list in $O(n \log n)$ time, $O(\log n)$ space.

Intuition: Merge sort fits linked lists — there's no random access for quick sort, but splitting at the middle and merging two sorted lists needs only pointer rewiring.

Key: same merge sort structure — slow/fast pointers to find mid, split, sort halves, merge.

```

class Solution {
    public ListNode sortList(ListNode head) {
        if (head == null || head.next == null) return head;
        ListNode slow = head, fast = head.next;
        while (fast != null && fast.next != null) {
            slow = slow.next; fast = fast.next.next;
        }
        ListNode mid = slow.next;
        slow.next = null;           // split into [head, slow] and [mid, ...]
        ListNode left = sortList(head);
        ListNode right = sortList(mid);
        return merge(left, right);
    }

    private ListNode merge(ListNode a, ListNode b) {
        ListNode dummy = new ListNode(0), current = dummy;
        while (a != null && b != null) {
            if (a.val <= b.val) {
                current.next = a;
                a = a.next;
            } else {
                current.next = b;
                b = b.next;
            }
            current = current.next;
        }
        current.next = (a != null) ? a : b;
        return dummy.next;
    }
}

```

Time $O(n \log n)$ | **Space** $O(\log n)$

#179 Largest Number

Description: Given a list of non-negative integers, arrange them to form the largest number.

Intuition: Order two numbers by which concatenation is bigger ($b+a$ vs $a+b$) — this pairwise rule sorts the whole list into the largest possible string.

Key: custom comparator — between a and b , prefer whichever concatenation $a+b$ vs $b+a$ is lexicographically larger.

```

class Solution {
    public String largestNumber(int[] nums) {
        String[] strs = new String[nums.length];
        for (int i = 0; i < nums.length; i++) strs[i] = String.valueOf(nums[i]);
        Arrays.sort(strs, (a, b) -> (b + a).compareTo(a + b));
        if (strs[0].equals("0")) return "0";    // all zeros
        return String.join("", strs);
    }
}

```

Time $O(n \log n)$ | **Space** $O(n)$

#315 Count of Smaller Numbers After Self

Description: For each element, count how many elements to its right are smaller.

Intuition: During a merge, whenever a left-half element is placed, every right-half element already emitted is both smaller and to its right — so the merge counts the smaller-after-self pairs for free.

Key: same merge sort template but track original indices. When element from left side is placed, $j - mid$ elements from the right side that have already been placed are all smaller and to its right.


```

class Solution {
    int[] result, indices;

    public List<Integer> countSmaller(int[] nums) {
        int n = nums.length;
        result = new int[n];
        indices = new int[n];
        for (int i = 0; i < n; i++) indices[i] = i;
        mergeSort(nums, 0, n, new int[n]);
        List<Integer> output = new ArrayList<>();
        for (int r : result) output.add(r);
        return output;
    }

    private void mergeSort(int[] nums, int start, int end, int[] temp) {
        if (end - start <= 1) return;
        int mid = start + (end - start) / 2;
        mergeSort(nums, start, mid, temp);
        mergeSort(nums, mid, end, temp);
        merge(nums, start, mid, end, temp);
    }

    private void merge(int[] nums, int start, int mid, int end, int[] temp) {
        int i = start, j = mid, k = start;
        while (i < mid || j < end) {
            if (j >= end || (i < mid && nums[indices[i]] <= nums[indices[j]])) {
                result[indices[i]] += j - mid; // j-mid right-side elements already placed
                temp[k++] = indices[i++];
            } else {
                temp[k++] = indices[j++];
            }
        }
        for (i = start; i < end; i++) indices[i] = temp[i];
    }
}

```

Time $O(n \log n)$ | **Space** $O(n)$

#56 Merge Intervals

Description: Merge all overlapping intervals. Return array of non-overlapping intervals.

Intuition: After sorting by start, overlaps can only be with the most recent interval, so one scan either extends its end or appends a new one.

Key: sort by start time; greedily extend the last interval's end when overlap found.

```

class Solution {
    public int[][] merge(int[][] intervals) {
        Arrays.sort(intervals, (a, b) -> a[0] - b[0]);
        List<int[]> result = new ArrayList<>();
        for (int[] iv : intervals) {
            if (result.isEmpty() || result.get(result.size() - 1)[1] < iv[0]) {
                result.add(iv);
            } else {
                result.get(result.size() - 1)[1] = Math.max(result.get(result.size() - 1)[1], iv[1]);
            }
        }
        return result.toArray(new int[0][]);
    }
}

```

Time $O(n \log n)$ | **Space** $O(n)$

Algorithm Chooser

Signal	Algorithm
"sort array", stable, O(n) space OK	Merge Sort
"sort array", in-place, O(log n) space	Quick Sort
"k-th largest/smallest"	Quick Select — O(n) avg
"k closest / k most frequent"	Quick Select with custom comparator
"sort linked list"	Merge Sort (quick sort needs random access)
"count inversions / smaller after self"	Merge Sort — count during merge step
"arrange to maximize/compare"	Custom Sort comparator
"merge overlapping intervals"	Sort by start + linear scan
integers in bounded range	Counting Sort — O(n + range)

Half-open Interval Cheat Sheet

```
mergeSort(nums, start, end) → sorts [start, end)
  mid = start + (end-start)/2
  left:  [start, mid)
  right: [mid, end)
  base:  end - start <= 1

quickSort(nums, start, end) → sorts [start, end)
  after partition: [start,i) < pivot, [i,k) == pivot, [k,end) > pivot
  recurse: (start, i) and (k, end)
```

#88 Merge Sorted Array

Description: Given two sorted arrays `nums1` (with capacity for `m+n`) and `nums2`, merge `nums2` into `nums1` in-place.

Variation: Fill from the END backwards using three pointers `i` (end of `nums1` valid data), `j` (end of `nums2`), `k` (end of merged result). This avoids the need to shift elements.

```
class Solution {
  public void merge(int[] nums1, int m, int[] nums2, int n) {
    int i = m - 1, j = n - 1, k = m + n - 1; // ← VARIATION: three pointers from END
    while (i >= 0 && j >= 0)
      nums1[k--] = (nums1[i] >= nums2[j]) ? nums1[i--] : nums2[j--];
    while (j >= 0) nums1[k--] = nums2[j--]; // ← copy remaining nums2
  }
}
```

Time O(m+n) | Space O(1)

#327 Count of Range Sum

Description: Given an integer array, count the number of range sums `sum(i,j)` that lie within `[lower, upper]` (inclusive).

Variation: Build prefix sum array. Then use modified merge sort — during the merge phase, for each element in the left half, use two sliding pointers `j` and `k` into the right half to count valid prefix sums (those where `arr[k] - arr[i]` falls in range). This gives O(n log n) vs O(n²) brute force.

```

class Solution {
    public int countRangeSum(int[] nums, int lower, int upper) {
        int n = nums.length;
        long[] prefix = new long[n + 1];
        for (int i = 0; i < n; i++) prefix[i+1] = prefix[i] + nums[i];
        return mergeCount(prefix, 0, n + 1, lower, upper);
    }

    private int mergeCount(long[] arr, int start, int end, int lower, int upper) {
        if (end - start <= 1) return 0;
        int mid = (start + end) / 2;
        int count = mergeCount(arr, start, mid, lower, upper)
            + mergeCount(arr, mid, end, lower, upper);
        int j = mid, k = mid;
        for (int i = start; i < mid; i++) {
            while (j < end && arr[j] - arr[i] < lower) j++; // ← VARIATION: sliding lower bound
            while (k < end && arr[k] - arr[i] <= upper) k++; // ← VARIATION: sliding upper bound
            count += k - j; // count valid sums for this i
        }
        // Merge phase (sort the two halves)
        long[] sorted = Arrays.copyOfRange(arr, start, end);
        Arrays.sort(sorted);
        System.arraycopy(sorted, 0, arr, start, sorted.length);
        return count;
    }
}

```

Time $O(n \log n)$ | **Space** $O(n)$

#164 Maximum Gap

Description: Given an unsorted array, return the maximum difference between successive elements in its sorted form. Must run in linear time.

Variation: bucket sort by pigeonhole principle. With n elements, the max gap is at least $\lceil \frac{\max - \min}{n-1} \rceil$. Place elements into buckets of that size; the max gap spans between one bucket's max and the next non-empty bucket's min.

```

class Solution {
    public int maximumGap(int[] nums) {
        int n = nums.length;
        if (n < 2) return 0;
        int min = Integer.MAX_VALUE, max = Integer.MIN_VALUE;
        for (int num : nums) { min = Math.min(min, num); max = Math.max(max, num); }
        if (min == max) return 0;
        int bucketSize = Math.max(1, (max - min) / (n - 1)); // ← VARIATION: pigeonhole bucket size
        int bucketCount = (max - min) / bucketSize + 1;
        int[] bucketMin = new int[bucketCount], bucketMax = new int[bucketCount];
        Arrays.fill(bucketMin, Integer.MAX_VALUE);
        Arrays.fill(bucketMax, Integer.MIN_VALUE);
        for (int num : nums) {
            int b = (num - min) / bucketSize;
            bucketMin[b] = Math.min(bucketMin[b], num);
            bucketMax[b] = Math.max(bucketMax[b], num);
        }
        int maxGap = 0, previousMax = min;
        for (int b = 0; b < bucketCount; b++) {
            if (bucketMin[b] == Integer.MAX_VALUE) continue; // empty bucket
            maxGap = Math.max(maxGap, bucketMin[b] - previousMax); // ← VARIATION: gap across boundary
            previousMax = bucketMax[b];
        }
        return maxGap;
    }
}

```

Time $O(n)$ | **Space** $O(n)$